

La storia di Internet, 1985 - '86

- Tra il 1985 ed il 1986 gli host Internet passano da 2.000 a 30.000
- Il TCP è disponibile nelle workstation e nei PC, così come nel computer portatile Compaq di recentissima introduzione.
- Ethernet è accettata come tecnologia per il cablaggio di edifici e campus
- Ciò comporta il diffondersi di termini come bridging, routing
- Compaiono le prime multinazionali operanti nel settore delle reti, quali IBM, Proteon, Synoptis, Banian, Cabletron, Wellfleet e CISCO

30/4/1986: l'Italia si connette ad Internet

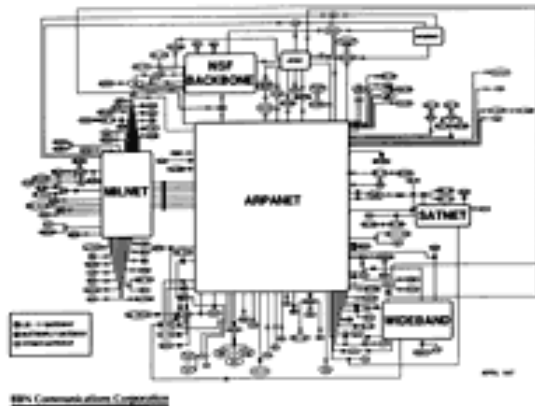
<https://www.raiplay.it/video/2016/05/Login-eaa2d582-187f-4ca7-8b64-ed9f9e6dc157.html>



STUDIO

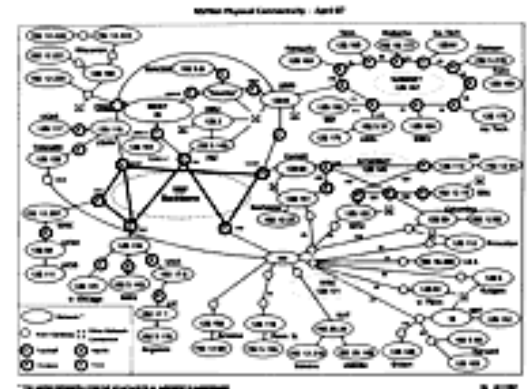
La storia di Internet, 1987

- Nel 1987 NFS intuisce la portata e il significato commerciale della velocità di crescita di Internet
- la rete NSF passa a linee T1 (1.55Mbps): la cosa ha un successo enorme e parte la proposta di migrare a linee T3 (45Mbps).
- Il numero di host passa a 10.000 e il numero di RFC supera quota 1.000
- Diventano serie le problematiche di Network Management



Internet Map, 1987

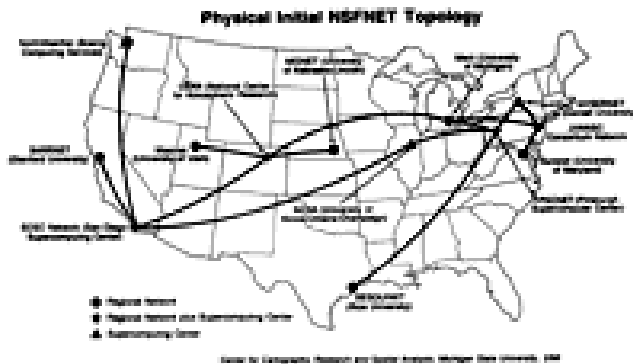
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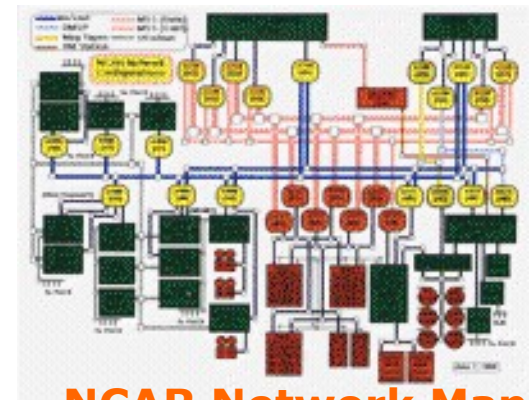
NSFNet Map, 1987

La storia di Internet, 1988

- NSF connette anche Canada, Danimarca, Finlandia, Francia, Norvegia e Svezia
- Nascono in California reti regionali quali Los Nettos e CERFNet
- **Fidonet**, il popolare sistema BBS, si connette ad Internet
- Nasce **Interop** evento-fiera per i produttori Internet
- Il primo WORM appare al CERT



NSFNet T-1 Backbone Map, 1988



NCAR Network Map, 1987

La storia di Internet, 1989

- Numero host: 80.000 in gennaio, 130.000 in luglio e 160.000 in novembre
- Si connettono Australia, Germania, Israele, Italia, Giappone, Messico, Olanda, Nuova Zelanda e Inghilterra.
- La velocità aumenta: le linee T3 (45M) diventano operative e ad Interop si osservano connessioni a 100Mbps in FDDI
- Le compagnie telefoniche attivano servizi a larga banda con il protocollo SMDS

La storia di Internet, 1989

- Vint Cerf e Bob Kahn organizzano il primo workshop sul **Gigabit**. Oltre 600 persone discutono su esperimenti da realizzare per una dorsale a 6 Gbps
-  **Tim Barners-Lee** al CERN pone il problema di una efficiente modalità di condivisione dei documenti riguardanti progetti e pubblicazioni scientifiche cercando di ottimizzare l'uso di carta e stampanti e proponendo il concetto di ipertesto: **nasce il World Wide Web!**

La storia di Internet, 1989

CERN DD/DC

Tim Berners-Lee, CERN/DD

Information Management: A Proposal

March 1989

Information Management: A Proposal

Abstract

This proposal concerns the management of general information about experiments and experiments at CERN. It discusses the problems of loss of information about complex evolving systems and derives a solution based on a distributed hypertext system.

Keywords: Hypertext, Computer conferencing, Document retrieval, Information management, Project management

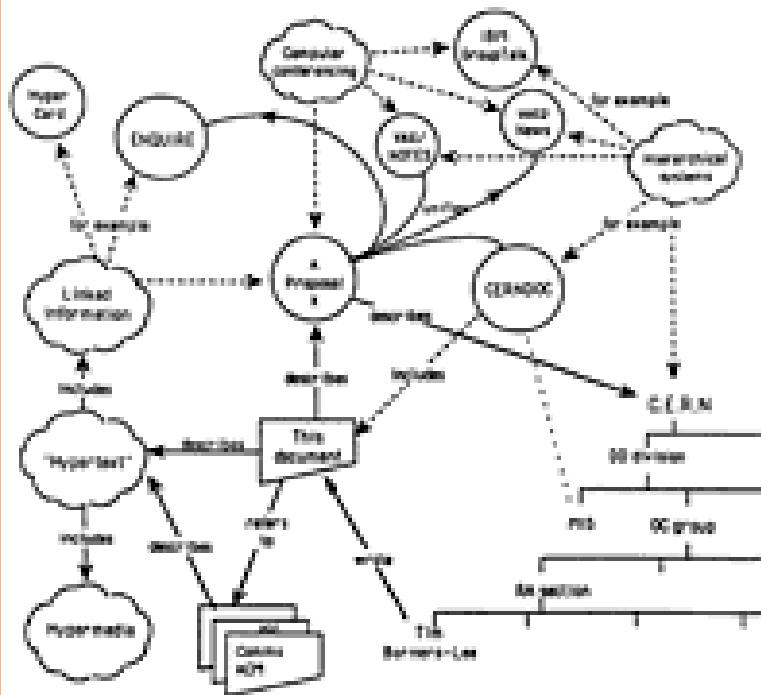


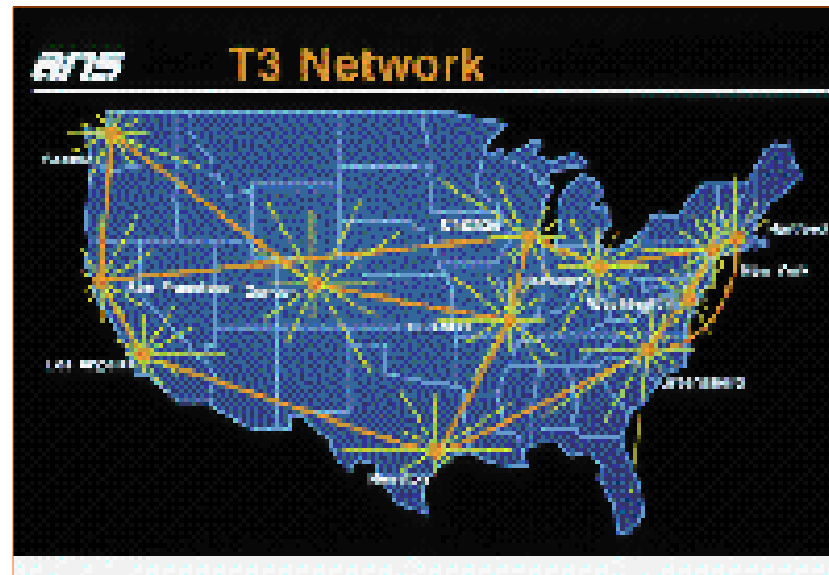
Diagramma
che descrive
l'ipertesto

La storia di Internet, 1990

- ARPANET viene chiusa: in venti anni è passata da 4 a 300.000 host. Tra le nazioni sono connesse: Argentina, Austria, Belgio, Brasile, Cile, Grecia, India, Irlanda, Sud Corea, Spagna e Svizzera
- Appaiono i tool di Network Information Retrieval: ARCHIE, Gopher, WAIS.
- 130 incidenti relativi a WORMS

La storia di Internet, 1991

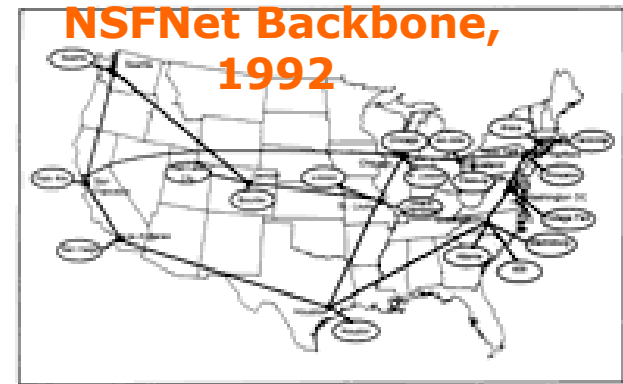
- NSF rimuove le restrizioni sull'uso commerciale della rete
- Nasce PGP, **Pretty Good Privacy**
- sono connesse più di 100 nazioni, 600.000 host e più di 5.000 reti separate



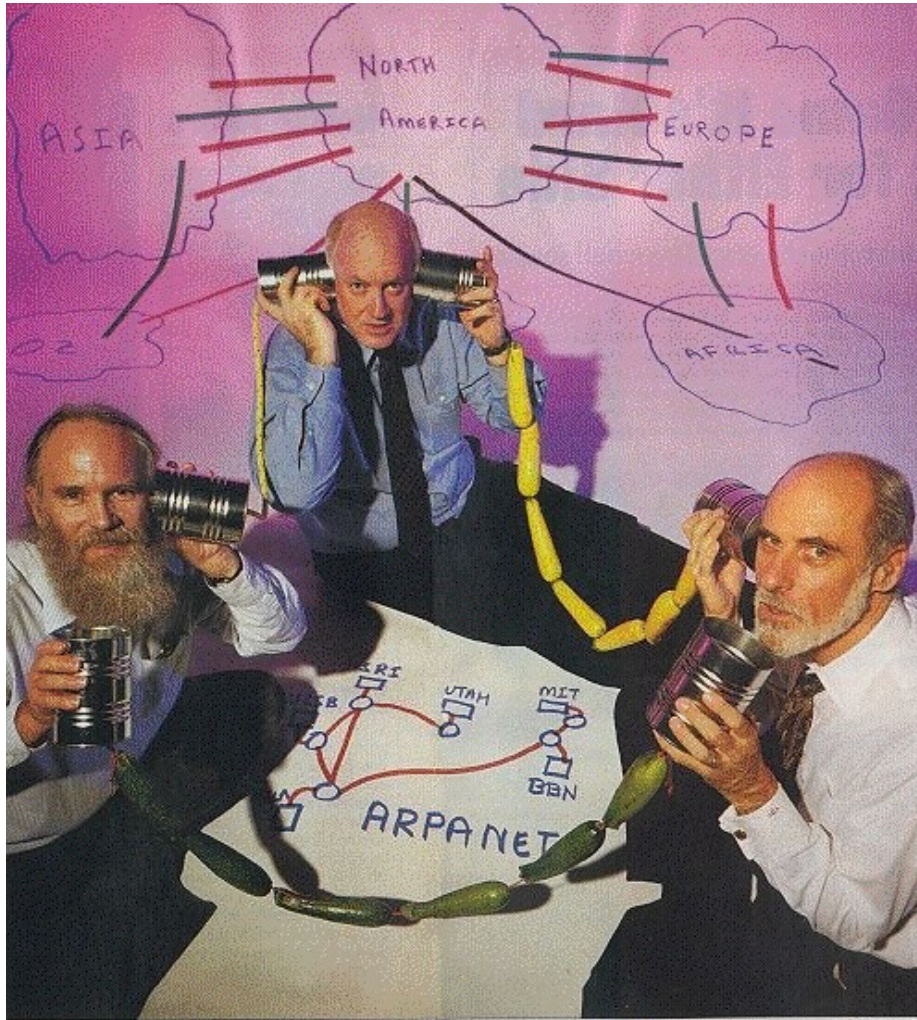
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La storia di Internet, 1992

- Nasce ISOC, che vede tra i fondatori Vint Cerf e Bob Kahn. L'Internet Activity Board diventa parte di ISOC.
- Più di 7.500 reti, 1.000.000 di host
- Il progetto MBONE consente l'uso di audio e video e sono possibili le prime videoconferenze in rete
- Presso NCSA Larry Smarr modifica la proposta di ipertesto di Tim Berners-Lee; nasce NCSA Mosaic. Una delle prime homepage mostrava la foto di Elvis e il link ad un suo brano...
- Marc Andreessen fonda con Jim Clark Netscape Inc
- **Il WWW esplode nella rete**



La storia di Internet, 1994



This photo was published in the August 8, 1994 issue of Newsweek and commemorates the 25th anniversary of the ARPANET. Jon Postel, Steve Crocker and I spent hours helping the photographer prepare for this shot.

Jon drew all the pictures, Steve and I strung the zucchini and the yellow squash. I think we must have collectively spent about 8 hours on this. Note that this network can't work - there is no mouth/ear link anywhere!!!

Such was the state of networking in the primitive 1960s...

Vint Cerf

La storia di Internet, 1995

- Resolution of FNC on October 24, 1995
- The Federal Networking Council (FNC) agrees that the following language reflects our definition of the term:
"**Internet**" refers to the global information system that
 - is logically linked together by a globally unique address space based on the Internet Protocol (IP) or its subsequent extensions/follow-ons;
 - is able to support communications using the Transmission Control Protocol/Internet Protocol (TCP/IP) suite or its subsequent extensions/follow-ons, and/or other IP-compatible protocols; and
 - provides, uses or makes accessible, either publicly or privately, high level services layered on the communications and related infrastructure described here in.

Domini Internet, 2004

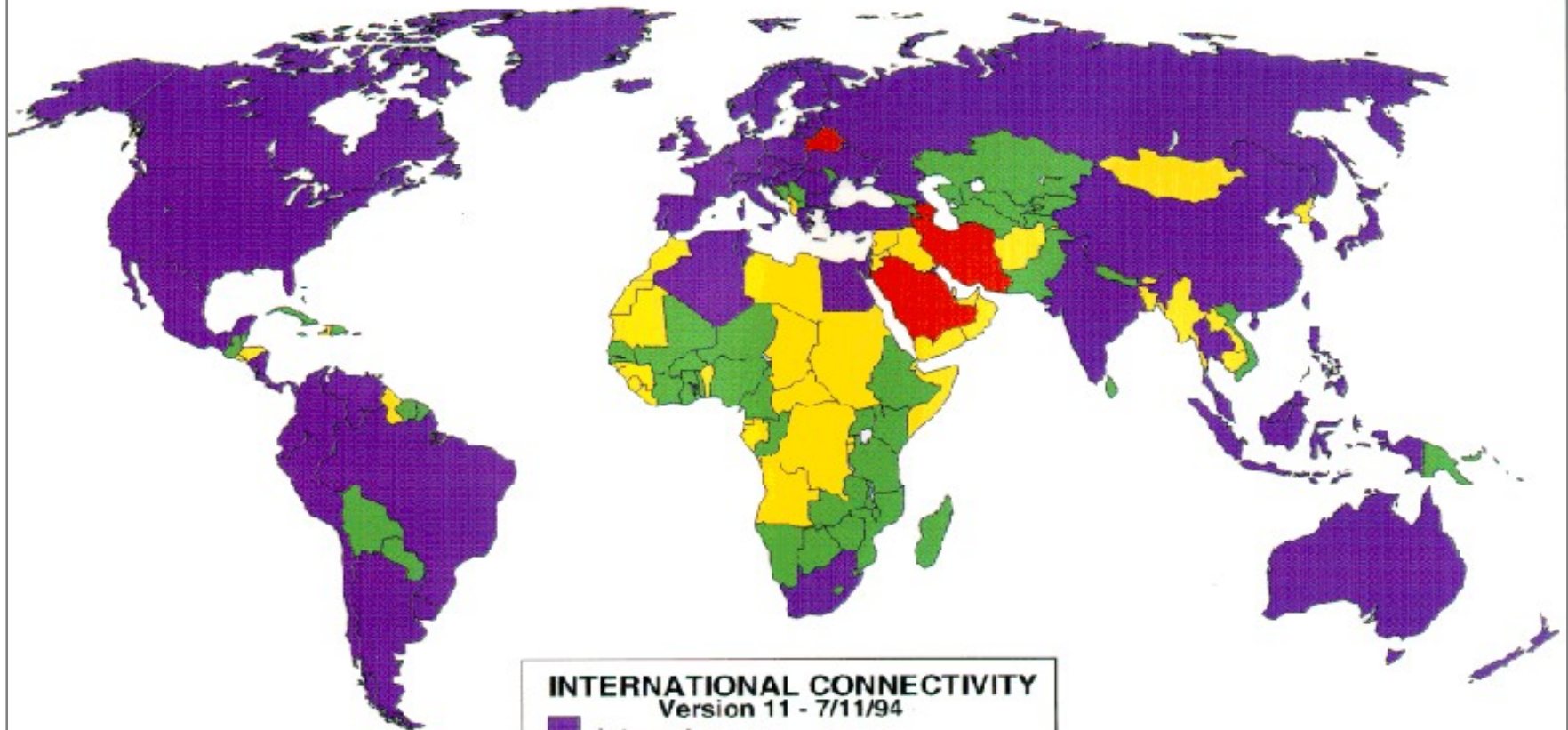
Total domains registered worldwide.....	62,9 milioni
Total .COM registered.....	22215530
INTERNATIONAL (COM)	22.215.530
INTERNATIONAL (EDU)	7.253
INTERNATIONAL (GOV)	1.322
INTERNATIONAL (NET)	3.851.052
INTERNATIONAL (ORG)	2.423.508
United Kingdom (CO.UK)	3.275.700

Solo nei primi tre mesi del 2004 sono stati assegnati nel mondo **4,1 milioni** di domini Internet

Domini .it: quota un milione

Connettività, 1994

Version 11 - 11 July 1994



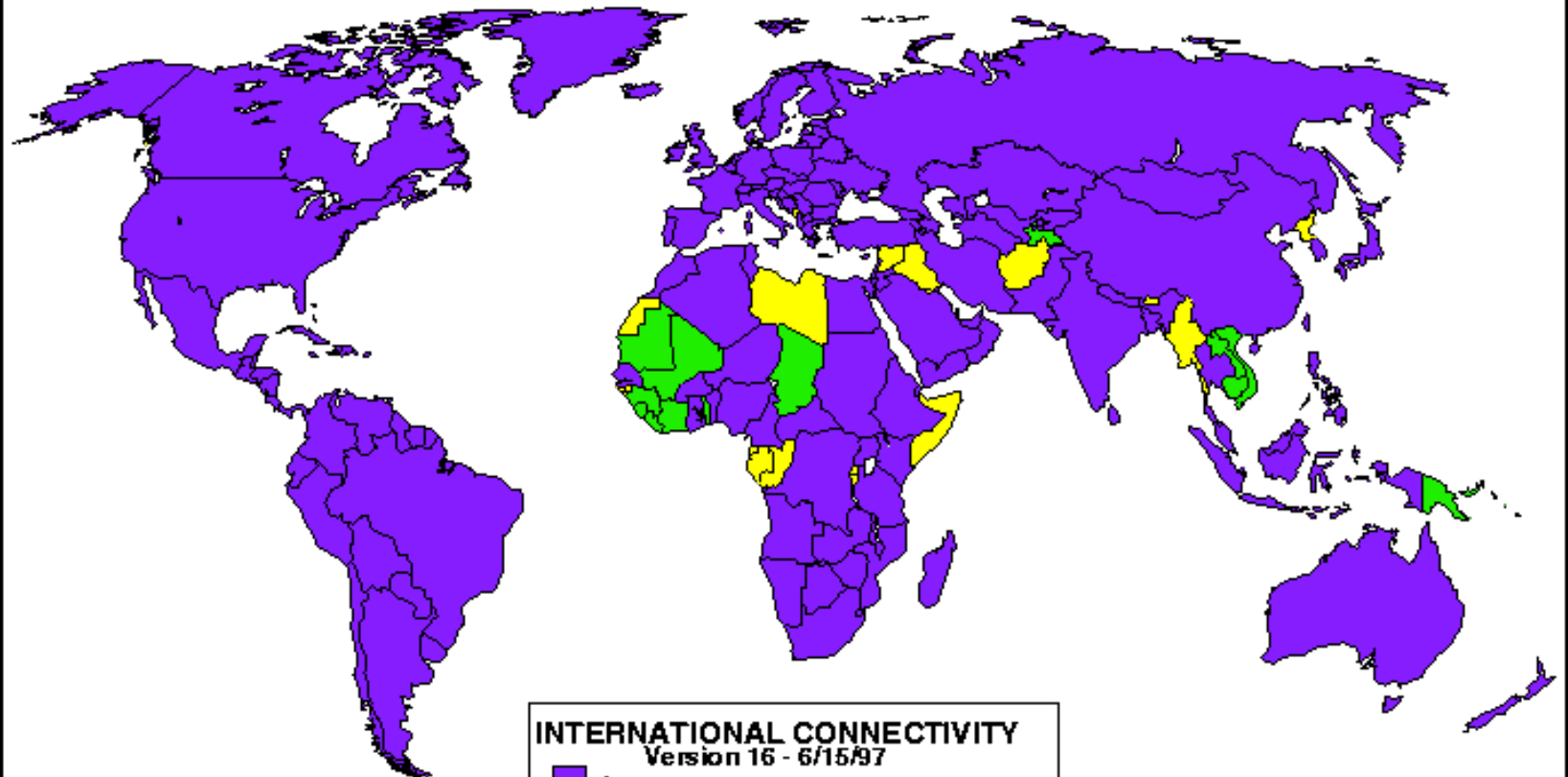
INTERNATIONAL CONNECTIVITY

Version 11 - 7/11/94

- Internet
- Bitnet but not Internet
- E-Mail Only (UUCP, FidoNet, or OSI)
- No Connectivity

Connettività (1997)

Version 16 - 16/06/1997



INTERNATIONAL CONNECTIVITY

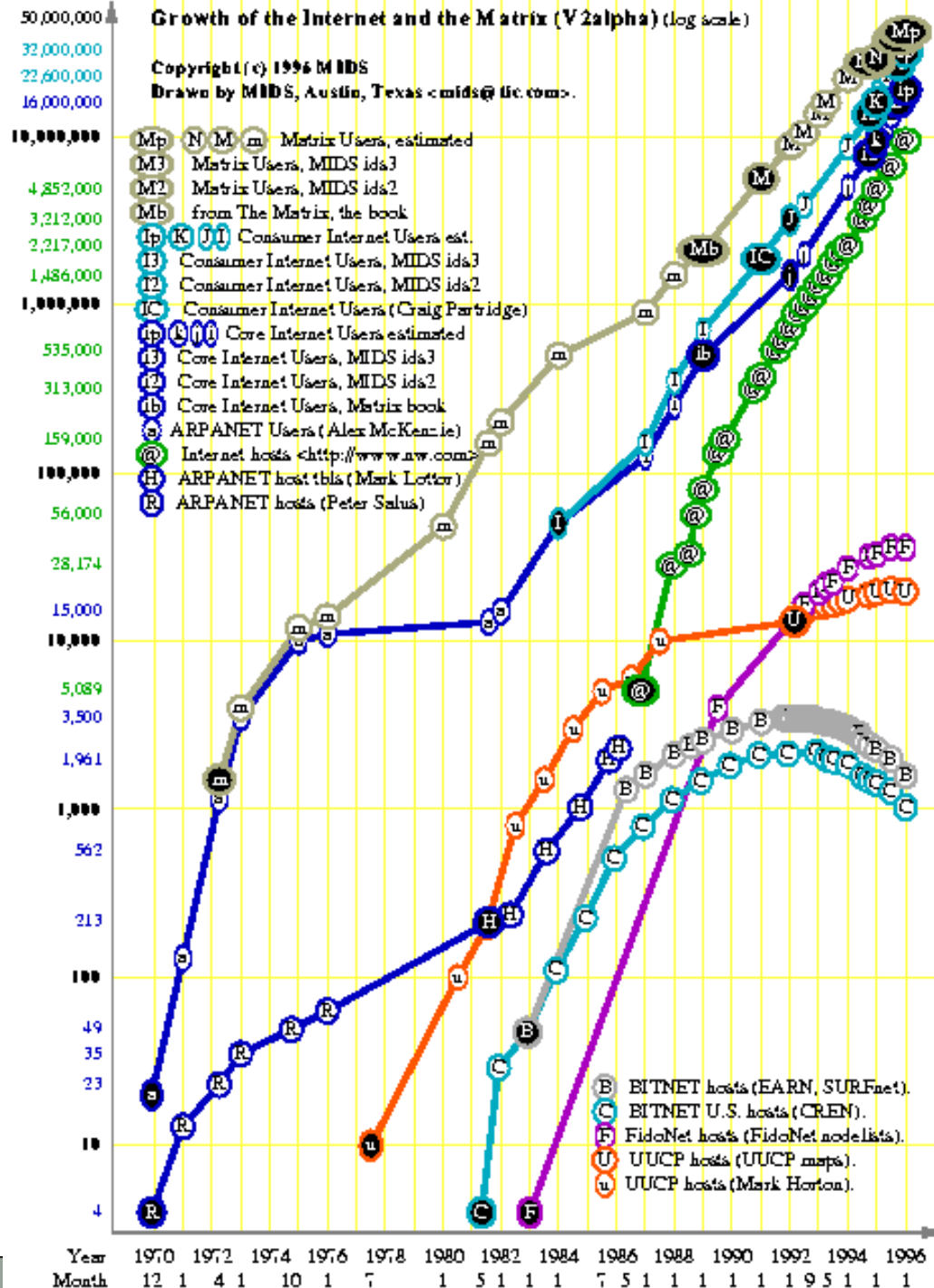
Version 16 - 6/15/97

- Internet
- Bitnet but not Internet
- EMail Only (UUCP, FidoNet)
- No Connectivity

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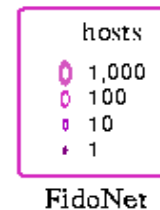
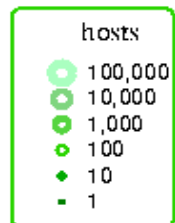
This map may be obtained via anonymous ftp
from <ftp://isi.wisc.edu/pub/connectivity/tibble/directory>

O. Gervasi



Matrix Survey
 by John Quartermann
<http://www.mids.org>

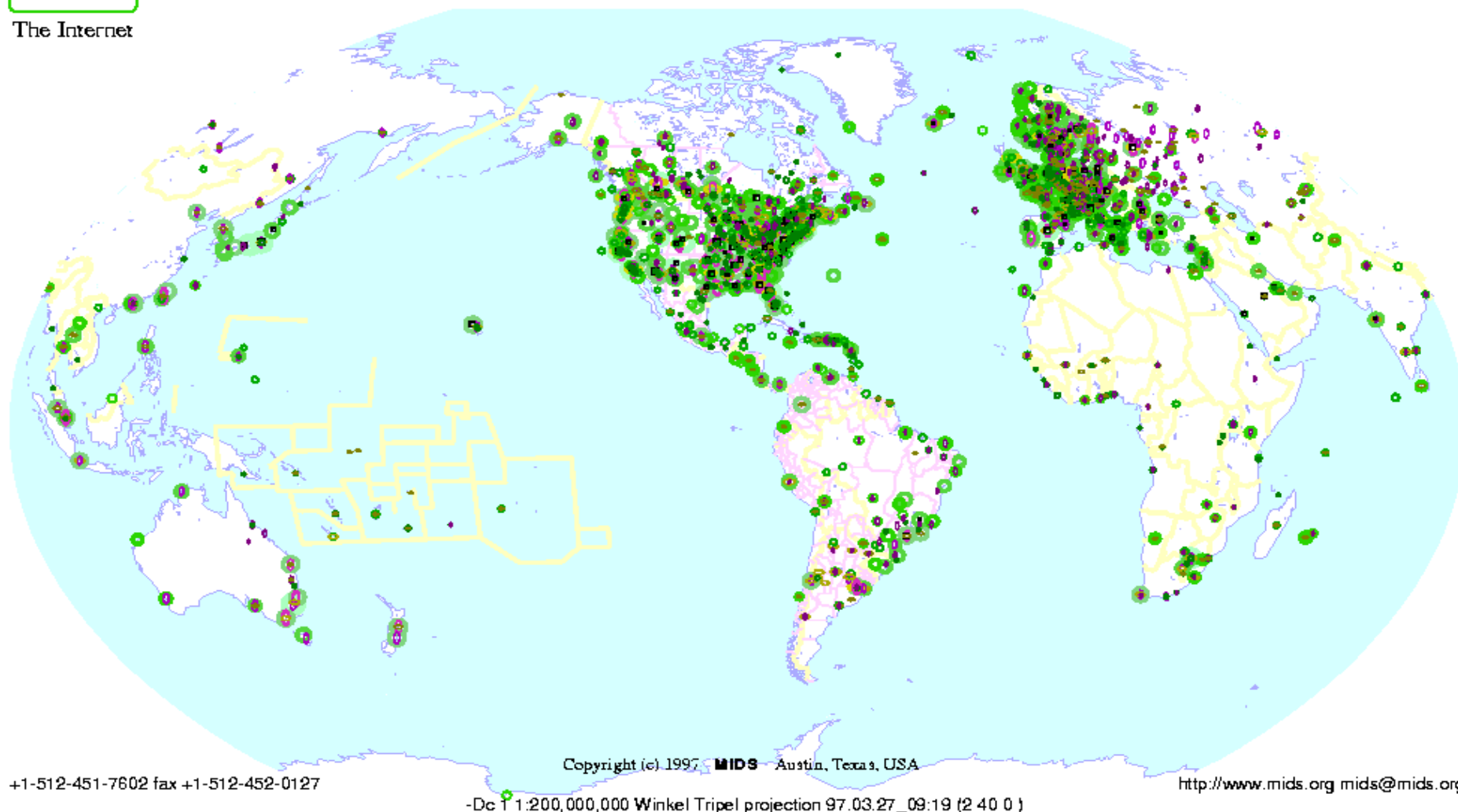
The Matrix Jan 1997



The World



The Internet



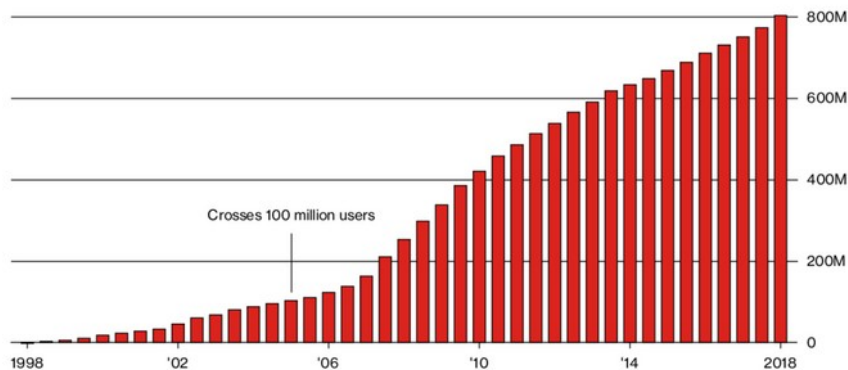
Statistiche pre-Covid19

- There are **4.1 billion Internet users** in the world as at December 2018. This is compared to 3.9 billion Internet users in mid 2018 and about 3.7 billion Internet users in late 2017.
- **Asia has the most Internet users of all continents** — accounting for 49 percent of all Internet users (down from about 50 percent in 2017 and up from about 48 percent mid 2018). Europe is a runner up with 16.8 percent of all Internet users.
- **China has the most Internet users of all country.** At over 802 million Internet users at the time of writing this, China currently accounts for almost 20 percent of Internet users worldwide. It is trailed by India, with over 500 million Internet users.

Web Supremacy

China internet users top combined population of Japan, Russia, Mexico and U.S.

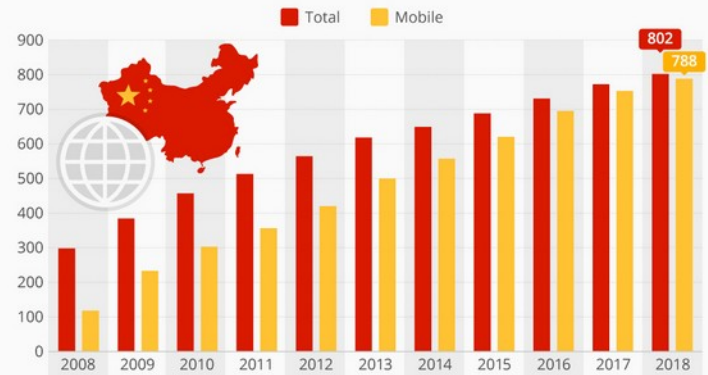
■ Number of internet users in China



Data: China Internet Network Information Center; graphic by Bloomberg Businessweek

98% Of Chinese Internet Users Are Mobile

Number of internet users in China (millions)



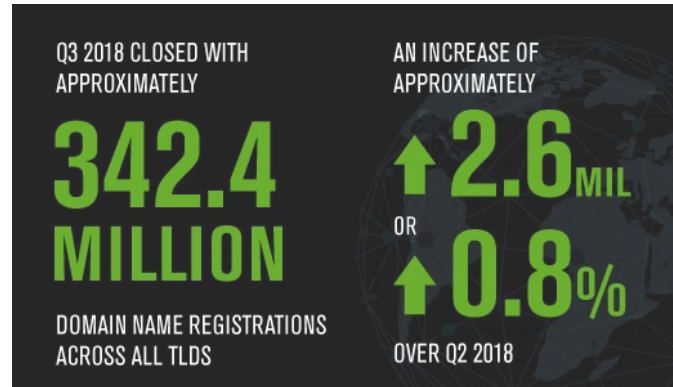
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CC BY SA

@StatistaCharts Source: CNNIC

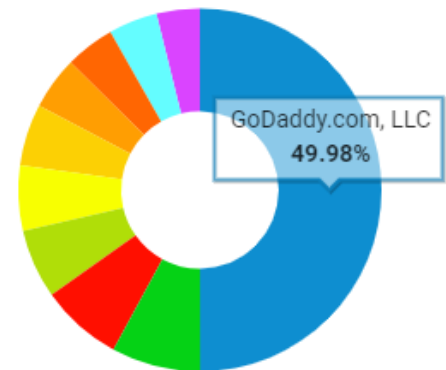
statista

Statistiche pre Covid19



Top registrars distribution

By number of registered domains



GoDaddy.com, LLC	49.98%
Tucows Domains Inc.	7.92%
NameCheap, Inc.	7.30%
eNom, LLC	6.17%
Network Solutions, LLC	5.82%
1&1 Internet SE	5.50%

Top TLDs distribution

By number of registered domains



.com	43.59%
.tk	6.65%
.net	4.68%
.de	4.04%
.uk	3.48%
.org	3.29%
.cn	3.21%
.ru	2.04%
.info	1.69%
Others	27.32%

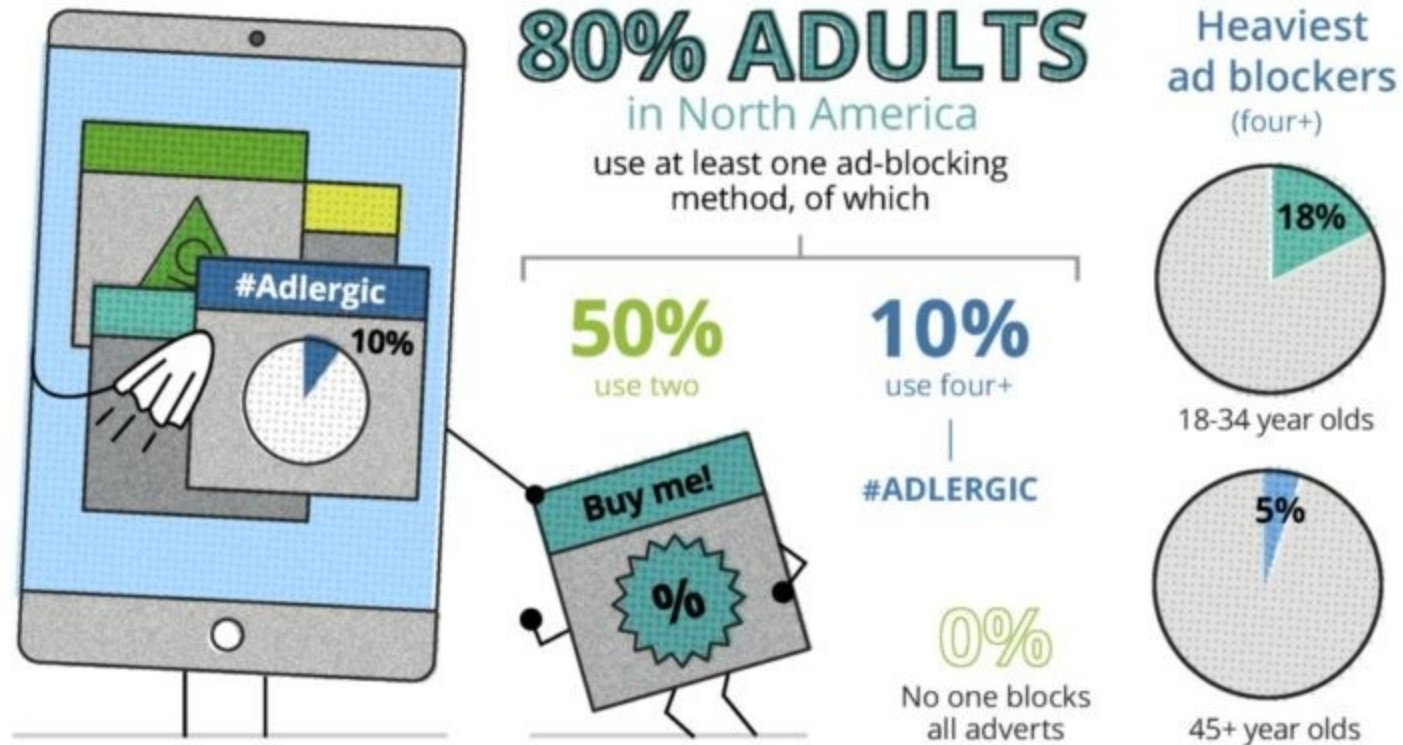
Statistiche pre-Covid19

- As at December 2018, there are approximately **1.94 billion** websites in the world.
- The world's first website was published on **August 6, 1991** by British physicist **Tim Berners-Lee**.
- **51.8 percent** of all Internet traffic comes from bots, while only **48.2 percent** of Internet traffic comes from humans.
- Google is the world's most visited website, followed by Youtube and Facebook. Chinese search engine Baidu is the world's fourth most visited website.
- With a 59.7 percent market share and about 22.6 million active sites, WordPress is the world's number one CMS. It hosts sites like *The New York Times*, *Forbes* and even the *Facebook Blog*.
- **Over 90,000 websites** are hacked every day.
- WordPress is the most hacked CMS — with 83 percent of hacked websites using the WordPress platform.

Statistiche pre-Covid19

Ad-Blocking Habits: Are Consumers 'Adlergic'?

Deloitte Global predicts that



Source: "Technology, Media and Telecommunications Predictions 2018," Deloitte Touche Tohmatsu Limited

Statistiche pre-Covid19

Top 10 US Companies*, Ranked by Retail Ecommerce Sales Share, 2018

% of US retail ecommerce sales

1. Amazon

49.1%

2. eBay

6.6%

3. Apple

3.9%

4. Walmart

3.7%

5. The Home Depot

1.5%

6. Best Buy

1.3%

7. Qurate Retail Group**

1.2%

8. Macy's

1.2%

9. Costco

1.2%

10. Wayfair

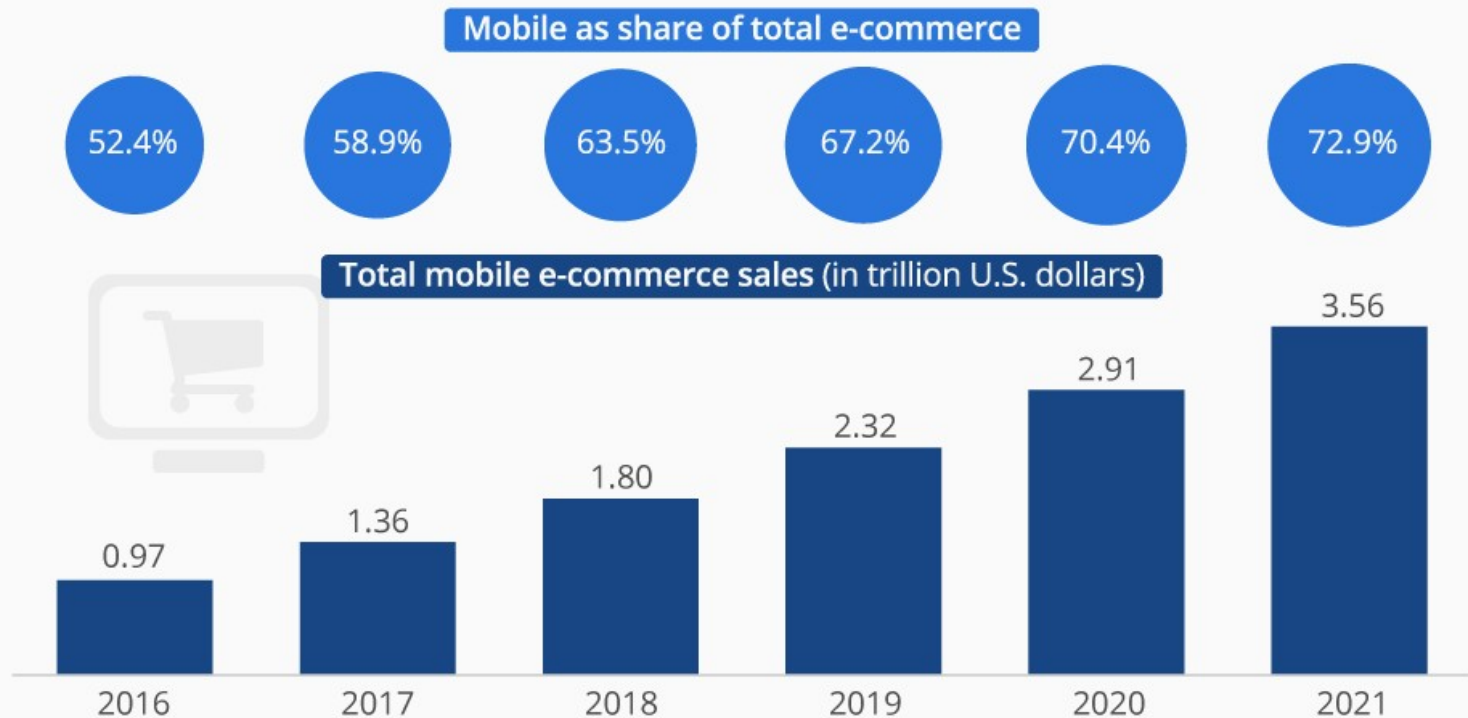
1.1%

Note: total US retail ecommerce sales=\$252.69 billion in 2018; top 10 companies' sales share=70.1% of total retail ecommerce in 2018; includes products or services ordered using the internet, regardless of the method of payment or fulfillment; excludes travel and event tickets; *excludes privately held companies; **includes ecommerce sales for QVC, HSN and zulily as of 2018; prior years included QVC only
Source: eMarketer, July 2018

Statistiche pre-Covid19

Mobile E-Commerce is up and Poised for Further Growth

Estimated mobile e-commerce sales worldwide



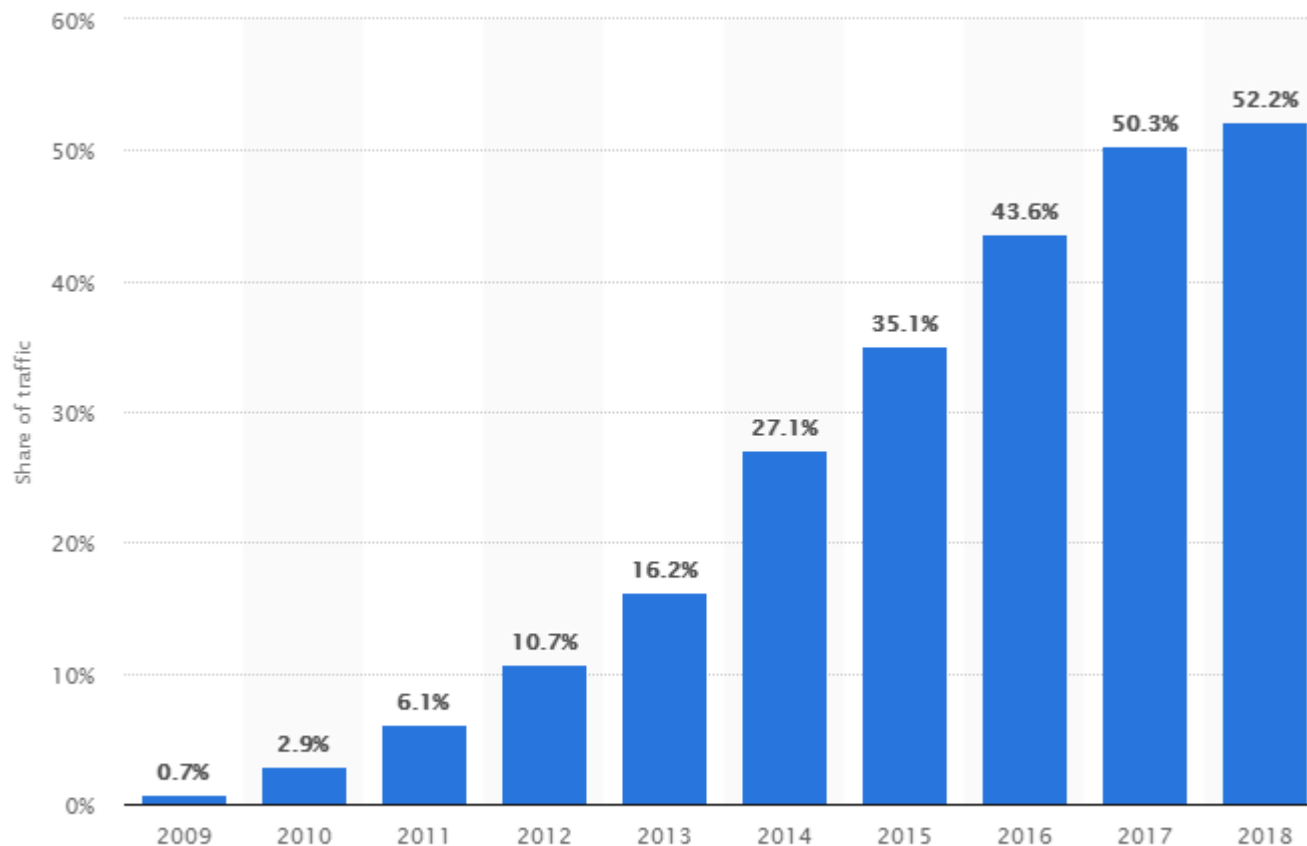
Includes products or services purchased via mobile devices (including tablets) regardless of the method of payment or fulfilment; excludes travel and event tickets

As of January 2018

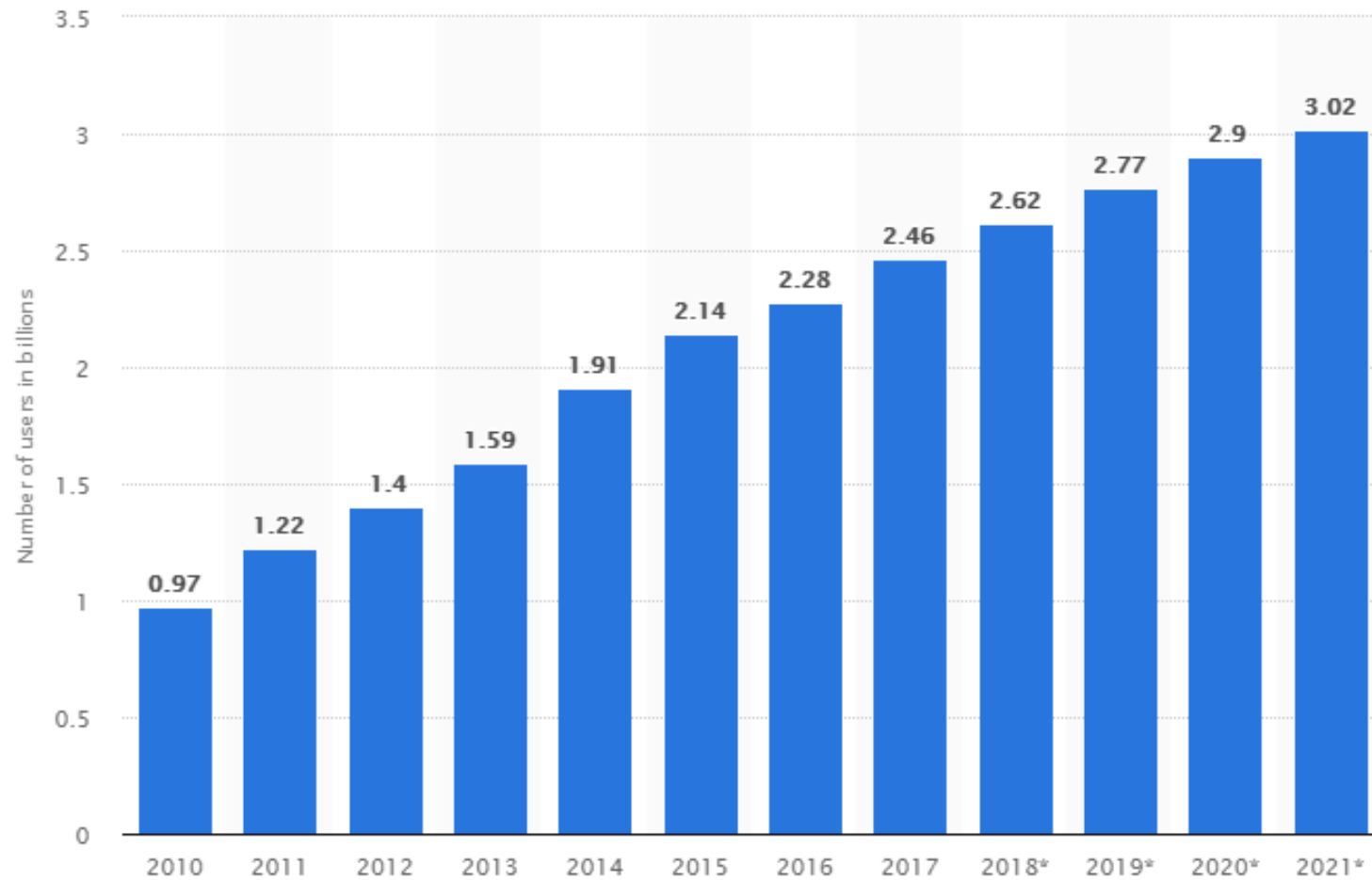
Source: eMarketer

Statistiche pre-Covid19

Percentage of all global web pages served to mobile |



Statistiche pre-Covid19



O. Gervasi

Statistiche 2022

- Utenti **5.1+ miliardi**, connessi per **6h 43'** ogni giorno
- 50 milioni in Italia, 70 milioni di cellulari
- Al 31/12/2021 erano 1.9 miliardi i siti web in Internet
- Google ora elabora 5.6 miliardi di query di ricerca ogni giorno in tutto il mondo.
- Google Chrome è il browser più diffuso (64.5% del mercato globale)

Statistiche 2022

- Nella prima metà di 2021, Il 64% di tutto il traffico Internet era automatizzato (il 39% proveniva da bot cattivi e il 25% da bot buoni). Gli esseri umani hanno rappresentato il restante 36%.
- Nel primo trimestre del 2021, i dispositivi mobili (esclusi i tablet) hanno generato il 54.8% del traffico globale del sito web.